

BRANDON WONG

New York, NY

☎ (917) 750-9908 ✉ brandonwong.0719@gmail.com 🔗 [linkedin.com/in/brandon0719](https://www.linkedin.com/in/brandon0719) 🐙 github.com/brandon0719 🌐 brandon0719.me

Education

Stony Brook University

Aug 2022 – May 2026

Bachelor of Science in Computer Science

Stony Brook, NY

- **Relevant Courses:** Data Structures & Algorithms, Software Engineering, Computer Networks, Operating Systems, Web Development, Web Security, Scripting Languages, Discrete Mathematics, Computer Systems Organization

Work Experience

Amazon Web Services (AWS)

May 2025 – Present

Software Development Engineer Intern

Seattle, WA

- Leveraged parallel Step Function workflows to coordinate **2 Lambda** functions to process **250,000+** computations concurrently, optimizing workflows by filtering out **96%** of unnecessary comparisons, reducing permission analysis processing time by **52%**, memory by **450%** enhancing scalability, and discover **2,000+** permission set similarities.
- Developing serverless architecture integrating with an existing metadata tool to fetch, process, and analyze **3,100+** permission sets across multiple AWS organizations using **6 AWS Lambda** functions implementing similarity algorithms to compare datasets to identify duplicates, and security vulnerabilities while storing recommendations in **Amazon S3**.
- Implemented **AWS EventBridge** for scheduled permission set analysis, reducing permission set and permission assignment technical debt by over **50%**, and enhancing security by analyzing permission drifts using **Python**.

Google

Feb 2025 – May 2025

Basta G-SWEP Fellow

Remote

- Engaged in **10 weeks** of pair programming and **mentorship** with a Google Software Engineer to strengthen problem-solving abilities, and enhance behavior and technical interview skills.

Stony Brook University

Aug 2023 – May 2025

Undergraduate Researcher

Stony Brook, NY

- Programmed and deployed a **JavaScript**-based web application aggregating health data from **4 APIs**, visualized through **8** dynamic charts and heatmaps using Plotly.js, and streamlined medical research workflows by creating a tool to fetch **10+** relevant articles from Google Scholar API.
- Identified key health trends by analyzing **40+** datasets with **R** and **JavaScript**, addressing challenges across **127** districts, and presented actionable insights and data visualizations to influence the direction of public health strategies.

Projects

Smart Parking System (P4SBU) | *Website* | *Source Code*

React | Node.js | PostgreSQL

- Collaborated with **3** peers to deploy a scalable web app enhancing parking experiences of **20,000+** Stony Brook faculty, students, and visitors, incorporating **MapBox API** for an interactive campus map with wayfinding algorithm based parking suggestions, a user-profile-driven reservation system, and an admin dashboard for oversight.

Phreddit | *Source Code*

React | Node.js | MongoDB

- Engineered a full-stack platform supporting **1000+** users using **CRUD** functionality to facilitate core features including community creation, threaded commenting, voting, context search, user profiles, and a reputation system, post creation, and engage in threaded discussions enhancing user engagement across the platform.

Leadership / Extracurricular

ChinaBlue

Aug 2023 – Present

Treasurer (2025–2026), Fundraising Head (2024–2025)

Stony Brook University

- Managed a **\$10,000+** budget and led weekly cabinet meetings to coordinate and execute cultural events.
- Led bi-weekly meetings to plan and spearhead 4 fundraising initiatives a semester, achieving a **210%** profit margin.

Technical Skills

Languages: Java, Python, HTML5, CSS, JavaScript, TypeScript, C

Developer Tools: AWS, Postman, Git, Agile, CI/CD

Libraries/Frameworks: NodeJS, ReactJS, PostgreSQL, MongoDB